# THE HALTING PROBLEM OF THE MACHINE $1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_---0LE$: A CONDITIONAL NON-HALTING THEOREM AND THE RECORD-TIMETABLE OBSTRUCTION

ABSTRACT. We give a complete structural analysis of the 6-state, 2-symbol Turing machine $M =$ `1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_---_0LE` and prove, conditional on a single precisely-stated computational assumption (master closure, Assumption M), that it never halts from the blank tape; unconditionally, non-halting holds over all runs whose round classes stay in the exhaustively verified range ($h \leq 2000$, sampled to $1.2 \cdot 10^7$) — a doubly-exponential horizon whose step-count scaling is labeled as empirical (Proposition on the envelope). The analysis proceeds by a macroscopic phase calculus between visits of state $D$ to blank cells, a closed family of recurrent anchors, a halving-chain prologue, and a *round law* proved by strong induction on the 2-adic class $h = \nu_2(\alpha + 1)$ of the leading counter, via a self-similar round decomposition (halving cascade, universal ladder, master segment). Magnitude decisions are handled unconditionally by a dominance inequality (stated as Lemma D of the companion derivation paper); the halting branch is structurally unreachable for $h \geq 82$ and perturbation-robust; $\Delta t(h) \in [2, 9]$ on all verified classes. The single remaining assumption is shown to be equivalent, by a finite-state reduction, to the continuation of a record timetable of the autonomous 2-adic walk, whose 32-bit slice is certified acyclic by a finite experiment (with non-eventual-periodicity beyond the slice stated as a conjecture) — the precise, irreducible core of the problem. Two further theorems, unconditional with respect to Assumption M, are proved along the way: a mod-3 ledger — from step 54 on, every phase configuration of $M$ has run-sum divisible by 3, so an entire half-space of configurations (including the Conway-chain entry $S(3, 1, 3)$) is unconditionally unreachable and the re-entry and anchor congruences hold as theorems — and a halt equivalence: $M$ halts iff some S-even birth value $A = x + y/2 + 1$ is a power of two (both directions algebraic; empirically, 2,908 births in the first $10^7$ phase events, none a power of two), with per-depth raw-reproduced witnesses (through depth 3) showing that birth guards fail at every finite depth, stated at witnessed level. All other components are proved algebraically or verified exhaustively over declared finite domains; every claim is classified in a ledger (algebraic / exhaustive / assumption-dependent).

## 1. INTRODUCTION

Let $M$ denote the Turing machine with states $A, \ldots, F$ and transition table of Table 1, in the standard notation where, e.g., `1RB` means: write 1, move right, go to state $B$, and where the single undefined transition (state $F$ reading 0) is the halting transition. The machine is one of the remaining undecided instances of the 6-state Busy Beaver problem; to our knowledge no analysis of it has been published.

**Theorem 1.1** (Main, conditional form)**.** *$M$, started on the blank tape, never halts, conditional on Assumption 5.10 (master closure for all classes).*

**Proposition 1.2** (unconditional envelope, verified range; empirical status made explicit)**.** *Unconditionally, $M$ does not halt within any run whose round classes stay in the exhaustively verified range (all classes $h \leq 2000$; sampled agreement through $1.2 \cdot 10^7$) — in step-count terms, non-halting is proven over a horizon that is doubly exponential in the class index (below $2^{2^{2000}}$ in the record-climb heuristic).*

*Date*: October 2026.

| | read 0 | read 1 |
|---|---|---|
| $A$ | 1RB | 1LA |
| $B$ | 0RC | 0RD |
| $C$ | 1LC | 1LA |
| $D$ | 1RE | 1RD |
| $E$ | 1LF | 0LA |
| $F$ | HALT | 0LE |

TABLE 1. The transition table of $M$.

*Status of the envelope.* The finite part is unconditional: over the verified range the round table is single-path and carries no halting block (Theorem 5.3, computational components of §9); the anchors climb by $n' \geq n + 6$ per exit, and the record timetable of the residual climbs by a measured factor $\approx 4$ per class level, so the first arrival of class $h$ requires a number of rounds growing like $4^h$, each of supercubic cost in the anchor index — hence the doubly-exponential horizon. The record climb, however, is a *measured* law of the autonomous walk (Theorem 9.1), not an unconditional theorem; we therefore state the envelope as a proposition whose step-count translation carries this explicitly labeled empirical component. A paper proof of an explicit lower bound $T(h) \geq 2^{c \cdot 2^h}$ (assembling $A_j \leq \tau_j^2$ with the band $\Delta\alpha \leq 2h+6$) is left open. Independently of the class range, the shape of the unconditional envelope is sharpened by Theorems 4.3 and 4.6: every configuration with $\Sigma \not\equiv 0 \pmod 3$ is unreachable from step 54 on (an entire half-space, with no appeal to any verified class range), and "$M$ never halts" is equivalent to "no power-of-two S-even birth ever occurs" — the verified domain of the latter equivalence being the first $10^7$ phase events (2,908 births, none a power of two; grade B). □

The proof occupies Sections 2–8. Its skeleton:

(1) (§2) A phase calculus for the instants when the head, in state $D$, reads a 0. Between two such instants the machine executes a fully determined finite block of elementary steps, and the possible tape shapes at these instants form a finite grammar with ten transition rules (nine non-halting event edges and the halting block; Lemma 2.3). The unique halting path passes through a single grammar configuration $G(3, \cdot)$ (Lemma 2.4).
(2) (§3) The blank-tape run reaches at step 54 an *anchor* $P(3)$, and every anchor $P(n)$ is carried, by an explicit halving chain of length $m = \nu_2(n+3)$, into a standard family $\Phi(w, m)$, $n + 3 = 2^m w$, $w$ odd (Theorem 3.4).
(3) (§5–5.2) The $\Phi$-trajectory visits *landmarks* $L(\alpha, t; s)$ indexed by a leading counter $\alpha \equiv 1 \pmod 6$ and a time parameter $t$; the evolution of $(\alpha, t)$ depends only on the class $h = \nu_2(\alpha + 1)$ (Theorem 5.3, the *round law*), proved by strong induction on $h$ via a halving cascade (Lemma 5.4), a universal ladder (Lemma 5.8), and a uniform master segment (Assumption 5.10, uniformity clause — Remark 5.11).
(4) (§6–8) Exhausted-tail configurations exit to the next anchor with $n' \geq n + 6$ and $n' \equiv 3 \pmod 6$ (Theorem 6.1), and no reachable configuration is the halting one (Theorem 7.1). The anchors therefore climb forever.

Throughout, tape contents are described by *runs* (maximal blocks of 1s) separated by *gaps* (blocks of 0s); all configurations live on the right-infinite half of the tape beyond a fixed left sentinel, which the machine never crosses (this follows from the left-moving transitions returning before the sentinel; it is confirmed by every invariant below).

## 2. The phase calculus

### 2.1. **Events and configurations.**

**Definition 2.1** (event)**.** An *event* is an instant at which the head, in state $D$, reads a 0. Let $\mathcal{C}$ denote the set of tape shapes (with head position) that can hold at an event.

**Definition 2.2** (grammar)**.** Write a configuration as a list of runs $(r_1, r_2, \dots, r_k)$, $k \geq 2$, with specified gaps between them and a marked head cell (the 0 being read). The following four types exhaust the configurations that arise on the orbits considered:

- $R(a, b)$: runs $(a, b)$, gaps [1], head on the fresh 0 *immediately right* of the right run;
- $S_k(x, y, z, \dots)$: $k \geq 3$ runs, gaps $[1, 2, 1, 2, \dots]$, head on the *left* 0 of the 2-gap between runs 2 and 3;
- $G_k(a, b, \dots)$: $k \geq 2$ runs, gaps $[3, 1, 1, \dots]$, head on the *middle* 0 of the 3-gap between runs 1 and 2;
- $L_k(r_1, r_2, r_3, \dots)$: $k \geq 3$ runs, gaps $[1, 1, 1, \dots]$, head on the single-0 gap between runs 2 and 3.

(Thus $L_2$ is identified with $R$: both have the head just right of the block.)

**Lemma 2.3** (phase rules)**.** *Between consecutive events, $M$ executes exactly the following transitions, with the stated exact step counts:*

| | | |
|---|---|---|
| *(R)* | $R(a,b) \to S_3(a{+}1,\ b{-}2,\ 1)$ | $2b+3, \quad b \geq 3;$ |
| *(S≥)* | $S_k(\dots, x, y, z, \dots) \to S_k(\dots, x{+}1,\ y{-}2,\ z{+}1, \dots)$ | $2y+3, \quad y \geq 3;$ |
| *(S2)* | $S_k(\dots, x, 2, z, \dots) \to G_{k-1}(\dots, x{+}1,\ z{+}1, \dots)$ | $7;$ |
| *(S1)* | $S_k(\dots, x, 1, z, \dots) \to L_{k-1}(\dots, 1,\ x{+}z{+}3, \dots)$ | $2x+z+14;$ |
| *(G≥)* | $G_k(a, b, \dots) \to S_{k+2}(1,\ a{-}3,\ 1,\ b{+}1, \dots)$ | $2a+3, \quad a \geq 4;$ |
| *(G3)* | $G_k(3, b, \dots) \to G_{k+1}(1,\ 1,\ b{+}1, \dots)$ | $9;$ |
| *(G2)* | $G_k(2, b, \dots) \to L_{k+1}(1,\ 3,\ b{+}1, \dots)$ | $16;$ |
| *(G1)* | $G_k(1, b, \dots) \to \mathit{HALT}$ | $6;$ |
| *(L≥)* | $L_k(r_1, r_2, r_3, \dots) \to L_k(r_1{+}1,\ r_2,\ r_3{-}1, \dots)$ | $2r_2+5, \quad r_3 \geq 2;$ |
| *(L1)* | $L_k(r_1, r_2, 1, \dots) \to R/S_{k-1}(r_1{+}1,\ r_2, \dots)$ | $2r_2+5.$ |

*In (S1) the result is $R$ when $k = 3$ and $L_{k-1}$ when $k \geq 4$; in (L1) it is $R$ when $k = 3$ and $S_{k-1}$ when $k \geq 4$. The halting block (G1) ends with $F$ reading 0; the block (G3) is an ordinary event block whose target $G(1, \cdot)$ halts in the next block.*

*Proof sketch.* Each rule is obtained by executing the transition table from the stated configuration until the next event. We illustrate rule (R) for $R(3, 3)$ (tape $\dots 1^3\,0\,1^3\,0^\infty$, head on the fresh 0, state $D$; the trace below is a step-level recording of the machine):

| $t$ | state/head | effect |
|---|---|---|
| 0 | $D$@7 | tentative write 1, right, $E$ |
| 1 | $E$@8 | write 1, left, $F$ |
| 2 | $F$@7 | *erase* (`0LE`), `left`, $E$ |
| 3 | $E$@6 | *erase* (`0LA`), `left`, $A$ |
| 4, 5 | $A$@5, $A$@4 | march left through the right run |
| 6 | $A$@3 | read the gap 0: *fill* it (`1RB`), `right`, $B$ |
| 7 | $B$@4 | write 0 (`0RD`), `right`, $D$ |
| 8 | $D$@5 | march right writing 1s |
| 9 | $D$@6 | next event: $S_3(4, 1, 1)$ |

The net effect is a five-phase mechanism: (i) $D$'s tentative extension of the right run is *undone* by the $E$–$F$–$E$–$A$ recoil; (ii) $A$ marches left across the right run; (iii) $A$ fills the inter-run gap with 1, growing the first run to length $a{+}1$; (iv) $B$ zeroes one cell of the old right run, creating the new 2-gap; (v) $D$ marches right, rebuilding, so that the old right run of length $b$ leaves a residue $b-2$ to the left of the new gap and a fresh run of length 1 to its right: $S_3(a+1, b-2, 1)$, at total cost $2b+3$. The remaining rules follow by the same kind of shuttling; each was derived stepwise from Table 1. Rule-level faithfulness is machine-checked in two complementary ways: (i) the rule checker `rulecheck.mjs` compares every rule against the raw machine on parameter grids ($a \in \{1, \ldots, 20\}$, $b \in \{2, 3, 5, 10\}$, tails of length 0–2): 564/564 rows agree exactly (this includes the boundary cases $y \in \{1, 2\}$, $a \in \{1, 2, 3\}$, and the halting case (G1)); (ii) trajectory-level comparison of the phase engine against the raw oracle over 956 anchor trajectories ($n = 2..957$): 955 agree field-by-field, including 16 halting anchors with exact halt times; the single exception is $R(2, 2)$, whose successor ($b = 2 < 3$) lies outside the declared grammar and is flagged, not mis-computed (direct simulation halts at +22). An earlier version of the table carried a folded (G3)$\to$ HALT row and an incorrect (G2) image $L(1, b{+}1, b{+}1)$; against that version the checker reported 531 ok / 21 bad, the bad rows being precisely (G2)/(G3) — the correction here follows the raw-machine side of that checker. □

**Lemma 2.4** (halting criterion)**.** *$M$ halts from an event configuration iff the phase trajectory reaches $G_k(1, b, \ldots)$ for some $b$ and tail; equivalently, iff it reaches a $G$-state with leading head $a \in \{1, 3\}$ (rule (G3) carries $G(3, \cdot)$ to $G(1, \cdot)$ in one block). On every orbit considered below, the leading heads of all $G$-states are $\geq 2$ and $\neq 3$, so no other rule can produce a halt.*

*Proof.* The only undefined transition of Table 1 is $F$ reading 0. An exhaustive check of the ten rules shows the state $F$ reads a 0 in the interior of an event block in exactly one grammar situation, namely the configuration $G(1, b, \ldots)$ (rule (G1)); in all other blocks $F$ reads 1s or the block ends at the next event first. Rule (G3) is the sole in-rule source of $G(1, \cdot)$. □

2.2. **Sweeps.** For efficiency we also use *swept* versions of the rules: consecutive applications of (S$\geq$) with the same $x$-growth collapse into one step with closed-form cost. If $y$ is odd, $j = (y-1)/2$ applications carry $(x, y, z) \mapsto (x + j,\ 1,\ z + j)$ at total cost $\sum_{i=0}^{j-1}(2(y-2i)+3)$; if $y$ is even, $j = (y-2)/2$ applications carry $(x, y, z) \mapsto (x + j,\ 2,\ z + j)$. These identities are elementary and are used silently below.

## 3. Anchors and the prologue

**Definition 3.1** (anchor)**.** For $n \geq 2$ let $P(n)$ denote the configuration $0^\infty 1^n 0 1^n 0^\infty$ with the head in state $D$ on the fresh 0 immediately right of the right run; in grammar terms $P(n) = R(n, n)$.

**Lemma 3.2** (entry)**.** *Started on the blank tape, $M$ reaches $P(3)$ at step 54.*

*Proof.* Direct simulation of the 54 steps: the machine builds the pattern $1^3 0 1^3$ by the $A$-$B$-$C$-$D$-$E$-$F$ loop and lands in state $D$ on the fresh 0. (Step-level trace in the repository; verified by two independent simulators.) □

**Definition 3.3** (the family $\Phi$)**.** For odd $w \geq 3$ and $m \geq 1$:

$$\Phi(w, m) \;=\; G\big(w-1,\ w-1,\ 2w,\ 4w,\ \ldots,\ 2^{m-1}w,\ 2^m w - 1\big).$$

**Theorem 3.4** (prologue)**.** *Let $n \equiv 3 \pmod 6$, $n \geq 3$ (this holds for every anchor on the orbit of Theorem 1.1; in particular $n$ is odd), and write $n + 3 = 2^m w$ with $w$ odd, $m \geq 1$. Then the phase trajectory from $P(n)$ reaches $\Phi(w, m)$; explicitly,*

$$R(n, n) \;\to\; R(1,\ 2n+2) \;\to\; G(n+2,\ n+1) \;\to^{m \text{ halvings}}\; \Phi(w, m),$$

*where the halving steps act on the two leading runs while the tail is carried through, and the leading heads follow the closed form*

$$a_i \;=\; 2^{m-i}\,w - 1, \qquad i = 0, 1, \ldots, m. \tag{1}$$

*Proof.* Since $n$ is odd, rule (R) gives $R(n, n) \to S_3(n+1,\, n-2,\, 1)$ with middle run $y = n-2$ *odd*; the sweep drives $y$ down to 1 in $j = (n-3)/2$ steps, and rule (S1) yields $R(1,\, x+z+3)$ with $x = n+1+j$, $z = 1+j$, i.e.

$$R(1,\, 2n+2).$$

Rule (R) on $b = 2n+2$ gives $S_3(2,\, 2n,\, 1)$ with $y = 2n$ *even*; the sweep drives $y$ down to 2 in $n-1$ steps, and rule (S2) yields

$$G\big(2+(n-1)+1,\; 1+(n-1)+1\big) \;=\; G(n+2,\, n+1).$$

Now let $G(a, b, \mathcal{R})$ be a halving state of the chain started at $G(n+2, n+1)$, i.e. $a+1 = 2^{m-i}w$ with $0 \le i \le m$. Since $w$ is odd, $a$ is odd exactly when $i < m$, so the halving map (rules (G≥) followed by the $L/S$ shuttles of §2, whose composite on the two leading runs is $(a, b) \mapsto \big((a-1)/2,\, (a-1)/2\big)$ with $b$ entering the accumulating geometric tail) applies for $i = 0, \ldots, m-1$. Starting from $a_0 + 1 = n+3 = 2^m w$ the heads satisfy (1), terminating at $a_m = w-1$ with accumulated tail $(2w, 4w, \ldots, 2^{m-1}w)$ and terminator $2^m w - 1$: exactly $\Phi(w, m)$. □

**Remark 3.5.** Theorem 3.4 is stated for $n \equiv 3 \pmod 6$, the congruence class of every anchor reached from $P(3)$ (Theorem 6.1, §8); for even $n$ the initial sweep branches differently and the chain above does not apply. The theorem is algebraic and holds for all $m \ge 1$; it was additionally machine-witnessed for all $n \equiv 3 \pmod 6$, $n \le 6003$ (1001 values), and for $m \in \{13, \ldots, 1200\}$, $w \in \{3, 9, 21\}$.

## 4. The mod-3 ledger and 2-adic births

This section contributes two theorems that are *unconditional* (they do not rely on Assumption 5.10) — a mod-3 ledger for the whole orbit (Theorem 4.3) and a halt equivalence in terms of power-of-two births (Theorem 4.6) — together with witnessed-level propositions on the pollution of birth guards (Propositions 4.7 and 4.8). Together they seal the mod-3 side unconditionally and condense the entire residual content of halt-exclusion into a single 2-adic statement about birth values.

**Definition 4.1** (ledger invariant $I_0$)**.** For a configuration $C$ let $\Sigma(C)$ denote the sum of its runs. $I_0$ is the set of configurations satisfying: (i) $\Sigma \equiv 0 \pmod 3$; (ii) domain: $R(a, b)$ with $b \ge 3$; $S(x, y, z, \overline{r})$ with $y \ge 1$; $G(a, b, \overline{r})$ with $a \ge 2$; $L(r_1, r_2, r_3, \overline{r})$ with $r_2 \ge 3$ and $r_3 \ge 1$; (iii) boundary guards BG1: $G(2, b, \overline{r})$ requires $b \ge 2$ and $b \equiv 2 \pmod 3$; BG2: $S(1, 2, z, \overline{r})$ requires $z \equiv 1 \pmod 3$. Both guards are *forced* by closure: BG2 because (S2) births $G(2, z+1)$, which must pass BG1; BG1 because the $\Sigma$-increment of the (G2) edge in the derivation engine's form is $b+1$ (at the raw level the increment is constantly +3; Remark 4.4). Under $\Sigma \wedge$BG2 the three-run class $S(1, 2, z)$ is empty ($z \equiv 0$ contradicts $z \equiv 1$); the class $S(1, 2, z, \overline{r})$ with $k \ge 4$ is nonempty by tail freedom, and the orbit runs precisely there ((G≥) at $a = 5$ births $S(1, 2, 1, b+1, \overline{r})$, $z = 1$).

**Lemma 4.2** (unique reachable value of $G(2, \cdot)$)**.** *The grammar-reachable configurations $G(2, b, \overline{r})$ are exactly those with $b = 2$ (including the geometric-tail family $\Phi(3, m)$).*

*Proof.* The only in-rule source of a $G$-state is (S2), $S(x, 2, z, \overline{r}) \to G(x+1, z+1, \overline{r})$; head 2 forces $x = 1$ at the sweep terminal. An $S$-entry with $x = 1$ can only come from (G≥) ((R) births $x = a+1 \ge 2$; (L1) births $x = r_1+1 \ge 2$; (S≥) sweeps increase $x$), i.e. $S(1, a-3, 1, b+1, \overline{r})$ with $z = 1$ fixed; reaching $y = 2$ with $x$ still 1 forces the sweep length $j = 0$, i.e. $a = 5$, yielding $S(1, 2, 1, b+1) \to G(2, 2, b+1, \overline{r})$. □

**Theorem 4.3** (mod-3 ledger; unconditional w.r.t. Assumption M)**.** *(a)* Closure. *Every member of $I_0$ steps to a member of $I_0$ under each of the ten rules of Lemma 2.3, the only exceptions being the halt forms $G(3, b, \overline{r})$ (classified, not violations). (b)* Seed. *The blank-tape orbit enters $R(3, 3) \in I_0$ at step 54 ($\Sigma = 6$). (c)* Conclusion. *Every phase (grammar) configuration of $M$ from step 54 on satisfies $\Sigma \equiv 0 \pmod 3$. Consequently every re-entry $R(1, b)$ has $b \equiv 2 \pmod 3$; every anchor $R(n, n)$ has $n \equiv 0 \pmod 3$; any halt must occur at a $\Sigma \equiv 0$ form $G(3, b, \overline{r})$; and every configuration with $\Sigma \not\equiv 0$ — including the Conway-chain entry $S(3, 1, 3)$ ($\Sigma = 7$) — is unreachable, an entire half-space. $I_0$ does not exclude halting (the halt forms $G(3, \cdot)$ can lie in $I_0$); the entire residual content of halt-exclusion lies on the 2-adic birth side (Theorem 4.6).*

[1]

Proof status (grade A$^-$, closure part): the closure is given by per-edge elementary affine identities plus guard transfers — (R): $(a{+}1) + (b{-}2) + 1 = a + b$; (S$\geq$) and (L$\geq$): translations; (S2): $(x{+}1) + (z{+}1) = x{+}z{+}2$; (S1), both branches: $+3$; (G$\geq$): $1 + (a{-}3) + 1 + (b{+}1) = a + b$; (G3): $\Delta = 0$; (G2): Remark 4.4; (L1), both branches: $+1$ — each hand-checkable, and re-verified by machine sweeps: the author's boundary enumeration of 219,398 $I_0$ member-steps plus $2^{40}$-scale random member-steps (the assertion total fluctuates by about $\pm 60$ with the random sampling; the deterministic core reproduces digit-for-digit), and an independent heterogeneous closure grid of 20,508 member-steps with run counts up to $k \leq 10$ and $a = 0$ probes. The reduction of closure to bounded run count is carried by locality (each rule rewrites only the first three runs; both guards read only the first three). Event-level raw fidelity was independently re-verified (all ten rules, boundary grids plus $2^{24}$-scale random); the seed is triple-sourced (two simulators plus an independent raw-machine parse). The orbit side is a redundant confirmation (the induction already yields the all-times statement): $10^7$ phase events with zero violations (406 anchors, all $n \equiv 0$; 1,485 re-entries, all $b \equiv 2$); an independent re-check of $5 \times 10^5$ events with zero violations (91 anchors, 332 re-entries); the trajectory library of 1,357,014 states with zero violations ($R$-family decomposition $825 = 547 + 150 + 128$, all $a + b \equiv 0$). The 112 halt forms $G(3, \cdot)$ mentioned above are an enumeration count of the closure grid, not an orbit observation (the orbit count of $G(3, \cdot)$ is 0).

**Remark 4.4** (raw fidelity domain of (G2))**.** Raw truth: the (G2) edge maps $G(2, b, \overline{r}) \to L(1, 3, b{+}1, \overline{r})$ for *all* $b \geq 2$ (raw-verified for $b \in \{2, 3, 5, 8, 11, 14\}$ with three tail shapes; verbatim the (G2) row of Lemma 2.3, whose rule checker 564/564 also covers (G2) at each $b$), with $\Sigma$-increment constantly $+3$ — so the mod-3 ledger is preserved by the true machine on all of $I_0$ *without appeal to BG1* (under the raw identity BG1 becomes superfluous; only the engine-level bookkeeping needs it). The derivation engine of the ledger analysis suite carries the simplified image $L(1, b{+}1, b{+}1, \overline{r})$, which agrees with the raw machine precisely at $b = 2$; by Lemma 4.2 this is the unique grammar-reachable value, so the distortion site $G(2, b \neq 2)$ is unreachable, and even if it were entered (started from an off-grammar configuration) the increment $+3$ would still preserve the ledger. Accordingly, the reading "ten rules = raw machine" must be qualified: the rule table is event-level raw-faithful on the grammar domain; on the off-grammar layer ($G(2, b \neq 2)$) the engine uses the simplified image, with no effect on the ledger.

**Definition 4.5** (S-even birth)**.** An event whose configuration is an $S(x, y, z, \overline{r})$ with $y$ even ($y \geq 2$) is called an *S-even birth* (the entry of that even sweep); its *birth value* is

$$A \;=\; x + \tfrac{y}{2} + 1.$$

[1] Granularity of the halt segment: the folded block (G3)+(G1) (15 steps total) contains, at $+9$ steps, the intermediate event $G(1, 1, b{+}1, \overline{r})$ — the target of (G3), outside the $G$-domain ($a \geq 2$) of $I_0$ but still with $\Sigma \equiv 0$ (the (G3) $\Sigma$-identity has $\Delta = 0$; spot-verified against the raw machine). The universal statement of Theorem 4.3(c) is thus pinned to *phase (grammar) configurations*.

**Theorem 4.6** (halt equivalence via power-of-two births)**.** *$M$ (started on the blank tape) halts if and only if some S-even birth value $A = x + y/2 + 1$ on its phase trajectory is a power of two; equivalently $A \in \{4, 8, 16, \dots\}$ (always $A \geq 3$, since $x \geq 1$ and $y \geq 2$; for $A = 3$ the sweep terminal has head 2 and feeds the non-halting (G2) channel).*

*Proof.* First, the *sweep invariant*: (S≥) preserves $A$, since $(x{+}1) + \frac{y-2}{2} + 1 = x + \frac{y}{2} + 1$; a birth sweep terminates at $S(x', 2, z')$ with $x' = x + \frac{y-2}{2}$, and (S2) lands in the $G$-state with head $x'{+}1 = A - 1$. ($\Leftarrow$) Let some birth have $A = 2^t$ ($t \geq 2$). Its sweep terminal $G$-state has head $2^t - 1$ (odd, $\geq 3$). For $t = 2$ this is already the halt form $G(3, \cdot)$. For $t \geq 3$ the head is $\geq 7$ and odd, and the halving composite of Theorem 3.4 (on the two leading runs, $(a, b) \mapsto \left(\frac{a-1}{2}, \frac{a-1}{2}\right)$ with the tail carried verbatim) applies at each step: the head chain runs $2^t - 1 \to 2^{t-1} - 1 \to \cdots \to 7 \to 3$ (every head on the chain has the form $2^s - 1 \geq 3$, never the value 5, never the $G(2, \cdot)$ channel), ending at the halt form $G(3, \cdot)$; by Lemma 2.4, $M$ halts. ($\Rightarrow$) Suppose $M$ halts. By Lemma 2.4 the trajectory reaches a $G$-state with head 3; the only in-rule source of $G(3, z{+}1, \overline{r})$ is (S2) acting on $S(2, 2, z, \overline{r})$. That $S$-state has $y = 2$ and lies inside some even sweep (or is itself its entry); the sweep invariant gives the birth value at that sweep's entry $A = 2 + \frac{2}{2} + 1 = 4 = 2^2$, a power of two. □

Proof status (grade A$^-$, both directions): both directions are elementary algebra, resting on rule faithfulness and the uniqueness of the halt form (heads $\in \{1, 3\}$ with only the (G3)/(G1) exits, raw-verified broadly); rule faithfulness is the B-grade check (rulecheck 564/564, 956 anchors) plus the independent event-level raw re-verification. Empirical companion (grade B): in the first $10^7$ phase events there are 2,908 S-even births, none a power of two; an independent re-check of $5 \times 10^5$ events finds 647 births, none. The entire content of Assumption M condenses to the single sentence "no power-of-two birth ever occurs".

**Proposition 4.7** (depth-1 birth guards are not closed)**.** *Let $I_1$ be $I_0$ together with same-shape algebraic guards at each birth site (odd part of the birth value $\geq 3$; the $f_2$ guard for $G(2, \cdot)$; $w \equiv 2 \pmod 3$ at $A = 3$ sites; exact forms in the* `invariant-casecheck` *suite). Then $I_1$ contains the entire real orbit (trajectory library:* 1,357,014 *states, zero violations), but $I_1$ is* not *closed: a closure search over the declared domain ($S/L$ tails of length $\leq 6$) yields* 1,255 *violations, all of a single class — an odd-$r_2$ $L$ arming, via (L1), its S-odd child, whose local guard is blind to the child's next-step birth.*

**Proposition 4.8** (per-depth pollution of birth guards; witnessed)**.** *For every tested depth $d \in \{1, 2, 3\}$ there is a configuration in $I_0$ whose pipeline has $d - 1$ S-even births that are not powers of two, followed by a power-of-two birth, and which truly halts:*

| *depth* | *witness* | *birth sequence* | *phase events* | *raw steps* |
|---|---|---|---|---|
| $d$=1 | $L[1, 3, 1, 3, 1]$ | $8, 4$ | *13* | *195* |
| $d$=1′ | $S[1, 1, 1, 1, 1, 1]$ | $8, 4$ | *15* | *230* |
| $d$=1″ | $G[4, 2, 7, 5]$ | $16, 8, 4$ | *33* | *728* |
| $d$=2 | $S[3, 2, 25, 43, 23, 12, 4, 2]$ | $5, 64$ | *152* | *10,313* |
| $d$=3 | $S[12, 28, 23, 42, 14, 22, 38, 31]$ | $27, 71, 128$ | *542* | *71,856* |

*All five witnesses were independently reproduced at the raw-machine level (event-by-event tape agreement with the phase configurations, raw halt confirmation, full chains inside $I_0$; additionally, the apparent non-halting witness $S[39, 34, 5, 30, 43, 19, 37]$ of the* 400*-step-capped search halts within* 442 *events under a larger cap — a cap artifact, not a counterexample). Moreover* residue-class refinement is impossible*: the bad births $\{2^t\}$ have residues mod* 12 *in $\{4, 8\}$, the very classes occupied by the orbit's own S-even heads (the $A$* mod 12 *histogram covers $\{0, \dots, 9\}$,*

*with* 9,282 *cases* $A \equiv 4$ *and* 396 *cases* $A \equiv 8$*); the intra-class split is exactly* $\nu_2$ *— irreducibly 2-adic. The entry edges of bad births form a* family of birth-entry edges*: (L1) dominant (the* 1,255 *violations of* $I_1$ *and the* $d = 2$ *witness both enter through it), with (R) also occurring (the* $d = 3$ *witness enters through (R)*$\to S_3$*); the pollution mechanism is the per-level translation of the (S1)/(L1) pipeline (Remark 4.9).*

**Remark 4.9** (structural translation for $d \geq 4$; not proved)**.** After each per-level repair (a depth-$d$ guard reads "the first $d$ births of the pipeline are non-powers-of-two"), an edge of the same shape reappears one level deeper in the next $L$-round: each repair translates the same-shaped pipeline by one level. Per-depth witnesses for every $d \geq 4$ are *expected* from this mechanism (random search stably produces fresh witnesses at each depth), but their per-$d$ existence is precisely the statement at issue and is not claimed here; "no finite guard depth closes" is accordingly stated at witnessed level (Proposition 4.8 plus this mechanism), not as a theorem.

**Remark 4.10** (relation to the no-precision conjecture)**.** The companion derivation paper states the no-precision conjecture for coarse templates (per-run residues at a fixed modulus plus linear inequalities); Proposition 4.8 does not prove it and does not change its status. What it supports, at witnessed level (grade B), is the strengthening direction beyond that language: residue classes and algebraic head-form guards alike fail at every finite guard depth. The forcing edges are not on the mod-3 side (Theorem 4.3 seals that side completely) but at the 2-adic birth sites: the entry edges form a family ((L1) dominant, (R) also occurring), with the (S1)/(L1) pipeline translating one level deeper per repair (Remark 4.9).

## 5. Landmarks and classes

Fix an odd *unit* $w \geq 3$ for the rest of the paper.

**Definition 5.1** (landmark)**.** For $\alpha \equiv 1 \pmod 6$, $\alpha \geq 7$, $t \geq 9$, and slack $s \geq 0$, let $E := 2^{t-3}w$ and

$$L(\alpha, t; s) \;=\; G\big(\alpha + 3 \cdot 2^{t-3}w,\ \alpha - 1 + 5 \cdot 2^{t-3}w,\ 2^t w,\ 2^{t+1}w,\ \ldots,\ 2^{t+s}w,\ 2^{t+s+1}w - 1\big).$$

Its *class* is $h = \nu_2(\alpha + 1)$.

**Lemma 5.2** (class parametrization)**.** *Fix* $h \geq 1$*. The integers* $\alpha \equiv 1 \pmod 6$*,* $\alpha \geq 7$ *with* $\nu_2(\alpha + 1) = h$ *are exactly*

$$\alpha \;=\; 2^{h+1}(3c + \rho_h) + 2^h - 1, \qquad c \geq 0,$$

*where* $\rho_h \in \{0, 1, 2\}$ *is the unique residue with* $(-1)^{h+1}\rho_h + (-1)^h \equiv 1 \pmod 3$*.*

*Proof.* $\alpha + 1 = 2^h\,\omega$ with $\omega$ odd forces $\omega = 2(3c + \rho) + 1$ with $\rho \in \{0, 1, 2\}$, $c \geq 0$, giving the shape. Modulo 3, using $2 \equiv -1$: the displayed congruence is the condition $\alpha \equiv 1$, which singles out one $\rho$ since the three values of $\rho$ give the three residues. □

**Theorem 5.3** (round law)**.** *There exist functions* $\Delta\alpha, \Delta t : \mathbb{N}_{\geq 1} \to \mathbb{N}$ *with* $\Delta\alpha(h) \equiv 0 \pmod 6$ *and* $\Delta t(h) \in [2, 9]$ *such that for every landmark* $L(\alpha, t; s)$ *of class* $h$*:*

1. *if* $s \geq \Delta t(h)$*, the phase trajectory returns to the landmark* $L\big(\alpha + \Delta\alpha(h),\ t + \Delta t(h);\ s - \Delta t(h)\big)$*, and no halting occurs;*
2. *if* $s < \Delta t(h)$*, the trajectory exits to the anchor* $P(n')$ *with* $n' = 2^{t+s+1}w + c$*,* $c \equiv 3 \pmod 6$*,* $c \geq 3$*.*

The proof runs through §5.1–5.3 by strong induction on $h$; it is completed in §5.3.

5.1. **The halving cascade.**

**Lemma 5.4** (cascade)**.** *Let $L(\alpha, t; s)$ have class $h$, write $\alpha+1 = 2^h\omega$ ($\omega$ odd), assume $t \geq h+4$, and put*

$$\Psi \;=\; \omega + 3 \cdot 2^{t-3-h} w.$$

*Then $\Psi$ is odd, and the first $h$ events of the round are the halving steps with leading heads*

$$a_i \;=\; 2^{h-i}\,\Psi - 1, \qquad i = 0, 1, \ldots, h,$$

*so that the $h$-th of them lands on the configuration*

$$S_0 \;=\; G\big(\Psi - 1,\ \Psi - 1,\ 2\Psi,\ 4\Psi,\ \ldots,\ 2^{h-1}\Psi,\ \underbrace{\alpha + 5E}_{\beta},\ 2^t w,\ \ldots,\ 2^{t+s+1} w - 1\big),$$

*i.e. on $\Phi(\Psi, h)$ with its final run $2^h\Psi{-}1$ replaced by the triple ($\beta$, $E$-geometric tail, $-1$ terminator), $\beta = \alpha + 5E$.*

*Proof.* The round opens with rule (G≥) on the leading head $a_0 = \alpha + 3 \cdot 2^{t-3} w$. Since $\alpha \equiv 1 \pmod 6$ and $3 \cdot 2^{t-3} w \equiv 0 \pmod 6$ ($t \geq 9$), $a_0$ is odd, and by the halving arithmetic of Theorem 3.4 the successive leading heads are $a_i = (a_0 - \delta_i)/2^i$-type where the affine bookkeeping gives $a_i + 1 = 2^{h-i}\Psi$: indeed $a_0 + 1 = \alpha + 1 + 3 \cdot 2^{t-3} w = 2^h\omega + 3 \cdot 2^{t-3} w = 2^h\Psi$ by definition of $\Psi$, and the halving map divides $a+1$ by 2 at each step while leaving the tail invariant, provided each intermediate head is odd — which holds because $a_i + 1 = 2^{h-i}\Psi$ with $\Psi$ odd. Meanwhile the displaced material accumulates as the geometric tail $(2\Psi, 4\Psi, \ldots)$ exactly as in the prologue. At $i = h$ the head is $\Psi - 1$ and the configuration is $S_0$ as stated; the run $\beta = \alpha + 5E$ arises because the second leading run of $L$ is $\alpha - 1 + 5E$ and the cascade transfers the increment $+2$ net. (Each such identity was verified symbolically and on $h = 1, \ldots, 60$ by exhaustive run-by-run comparison.) □

**Lemma 5.5** (entry to ladder scale)**.** *For $h \geq 12$, the trajectory from $S_0$ reaches, after reading only runs up to exponent $\leq 10 < h - 1$, the landmark-scale configuration $L(7, 9)$-type at unit $\Psi$ with the modified tail $(\beta, E\text{-tail}, -1)$ carried verbatim.*

*Proof.* From $S_0$ the machine consumes the $\Psi$-geometric runs in the fixed manner of a fresh $\Phi(\Psi, h)$-trajectory; the first landmark of any $\Phi$-trajectory is $L(7, 9)$-shaped (leading counter 7, time parameter 9), reached having read only the runs $2\Psi, \ldots, 2^9\Psi$. The modified tail sits entirely at exponents $\geq t > h + 4 \geq 16 > 10$, hence beyond the read window and carried verbatim. (Symbolic derivation; witnessed on $h = 12, \ldots, 60$.) □

5.2. **The universal ladder.**

**Definition 5.6** (ladder)**.** Define the ladder as the sequence $(A_j, \tau_j)_{j \geq 0}$ with $(A_0, \tau_0) = (7, 9)$ and

$$(A_{j+1}, \tau_{j+1}) \;=\; \big(A_j + \Delta\alpha(H_j),\ \tau_j + \Delta t(H_j)\big), \qquad H_j := \nu_2(A_j + 1),$$

where $\Delta\alpha, \Delta t$ are the functions of Theorem 5.3.

**Lemma 5.7** (autonomy)**.** *On any $\Phi$-trajectory (with any admissible modified tail), the successive landmarks follow the ladder: the $j$-th landmark is $L(A_j, \tau_j; \cdot)$ at the unit scale, with the tail parameters evolving independently. Consequently the sequence of classes visited depends only on $j$, not on the unit or the tail.*

*Proof.* By Theorem 5.3 the passage landmark→landmark changes $(\alpha, t)$ by $(\Delta\alpha(h), \Delta t(h))$ with $h$ the class of the current $\alpha$; the unit and tail enter only through the slack bookkeeping (which decides landmark-vs-exit, not the map itself). □

**Lemma 5.8** (margins and class bound)**.** *The ladder satisfies, for all computed positions:*

(1) $\tau_j \geq 2H_j + 16$ *for all* $j \geq 3$*;*
(2) $A_j \leq \tau_j^2$ *for all* $j \geq 8$.

*Moreover, if these inequalities hold up to position* $j$*, they persist:* $A_{j+1} \leq A_j + 2H_j + 6 \leq \tau_j^2 + 4\log_2 \tau_j + 6 \leq (\tau_j + 2)^2 \leq \tau_{j+1}^2$ *using* $\Delta t \geq 2$.

*Proof.* Induction as displayed, given the band bound $\Delta\alpha(H) \leq 2H+6$, which holds on all classes where the round law has been established (this is part of the induction of §5.3: the induction hypothesis at class $h$ includes the band bound for all classes $\leq h$). Items (1)–(2) were verified exhaustively on the first 200,000 ladder positions, which cover all $\tau_j \leq 10^6$. □

**Corollary 5.9** (well-foundedness)**.** *For* $h \geq 16$*, every ladder position with* $\tau_j \leq h - 1$ *satisfies* $H_j \leq 2\log_2 h < h$*: the ladder prefix traversed inside a round of class* $h$ *involves only rounds of class strictly smaller than* $h$.

5.3. **The master segment.** Fix $h \geq 16$ and a landmark $L(\alpha, t; s)$ of class $h$. By Lemmas 5.4–5.5 the round consists of: the cascade into $S_0$; a ladder prefix from $L(7, 9)$-scale up to the position $j_0$ = the last $j$ with $\tau_j \leq h-65$ (all of whose rounds have class $< h$ and are covered by the induction hypothesis); and the *master segment*: the events from the ladder literal $L(A_{j_0}, \tau_{j_0})$-scale to the end of the round.

**Computational Assumption 5.10** (master closure)**.** For every $h \geq 1$, the symbolic derivation from the ladder literal $(A_{j_0(h)}, \tau_{j_0(h)}, h)$ with the outer parametrization of Lemma 5.2 ($c \geq 0$, $u \geq 0$, $w = 6u + 3$) and $t \geq 2h + 16$ is *single-path*: every branch decision (parity and run-size comparisons) is forced uniformly over the entire domain, no split ever occurs, and the derivation terminates in $O(\mathrm{poly}(\log h))$ events at either the next real landmark (yielding $\Delta\alpha(h) \equiv 0 \bmod 6$, $\Delta t(h) \in [2, 9]$) or, if $s < \Delta t(h)$, at the anchor exit of Theorem 5.3(ii). Moreover the derivation is *$h$-uniform*: its complete form set depends on $h$ only through the ladder literals $(A_{j_0}, \tau_{j_0})$ and the exponent offsets of the modified tail (the uniformity clause, spelled out in Remark 5.11); this is what allows the finite audits below to be lifted from the verified classes to all $h$.

**Remark 5.11** (the uniformity clause and its evidence)**.** The $h$-uniformity clause of Assumption 5.10 states that the master derivation depends on $h$ only through the ladder literals $(A_{j_0}, \tau_{j_0})$ and the exponent offsets of the modified tail, whose margins grow monotonically with $h$ (the read window is bounded by $\Delta t + 3$ while the tail begins at exponent $t \geq 2h + 16$). Under this clause, any failure of the single-path property at some $h_* > 10^6$ would be witnessed by a violation of one of finitely many margin inequalities, each monotone in $h$ beyond the verified range; the clause was verified concretely on all classes $h \leq 2000$ (exhaustively) and on sampled classes through $1.2 \cdot 10^7$. It is part of the assumption, not a separate theorem: the monotonicity argument above is a structural observation over the verified derivation shapes, and the safety proof cites it as such (conditional on Assumption 5.10).

**Proposition 5.12** (evidence for Assumption 5.10)**.** *Assumption 5.10 (single-path clause and uniformity clause alike) was verified by exhaustive single-path symbolic derivation: (i) for all* $h = 16, \ldots, 1024$*, by direct agreement with the independent per-h direct engine (*1009/1009*); (ii) for* $h = 1025, \ldots, 2000$ *exhaustively and for sampled* $h$ *up to* $1.2 \cdot 10^7$ *(*976 *table rows plus on-the-fly ladder extensions, all single-path, all law-shape consistent); (iii) at eight classes* $h \in [1028, 1777]$ *against numeric ground truth (exact; the class* $h = 1538$ *among them realizes the rare value* $\Delta t = 9$*, correcting the earlier bound* $[2, 8]$ *— see the erratum in the dossier); (iv) at fresh classes* $h \in \{1823, 2048, 31337, 666667\}$ *— at* $h = 1823$ *with three-way agreement (master = direct engine = precomputed table) — and at* $h = 45678$ *via the published blind-test recipe, and — with the ladder recurrence computed on the fly rather than stored, cross-checked against*

*the stored ladder at $h = 1777$ — at the fresh classes $h \in \{9999991, 10^7, 10^7+3, 12345679\}$, each single-path with $\Delta\alpha/h \approx 1.3382$.*

*Proof of Theorem 5.3.* By strong induction on $h$. *Base $h \le 15$:* direct single-path symbolic derivations (the first 15 rows of the per-$h$ table). *Step $h \ge 16$:* the decomposition of §5.1–5.3 expresses the round as cascade (Lemma 5.4, algebraic) + entry (Lemma 5.5) + ladder prefix (rounds of class $< h$, each correct by the induction hypothesis and Corollary 5.9) + master segment (Assumption 5.10). The composite is single-path and ends in outcomes (i) or (ii). The band bound $\Delta\alpha(h) \le 2h+6$ required by Lemma 5.8 is carried along the same induction: it holds on the base, and the master outputs satisfy it on all verified classes ($\Delta\alpha/h \approx 1.338$ throughout, e.g. $\Delta\alpha(10^6) = 1338210$). □

## 6. Endgames

**Theorem 6.1** (anchor exit)**.** *Let $L(\alpha, t; s)$ of class $h$ occur on the $\Phi(w, m)$-trajectory started from the anchor $P(n)$ ($n + 3 = 2^m w$), and let $s < \Delta t(h)$. Then $s$ is the slack to the* terminator *run $2^m w - 1$ of $\Phi$, i.e. $t + s + 1 = m$, and the trajectory exits the $\Phi$-family at the anchor*

$$n' \;=\; 2^m\, w + c \;=\; (n+3) + c, \qquad c \equiv 3 \pmod 6,\ c \ge 3,$$

*so that $n' \ge n + 6$.*

*Proof.* Along a $\Phi$-trajectory the slack $s$ counts the geometric tail runs beyond the current read window; the trajectory ends in an exit exactly when the read window of the current round overruns the terminator $2^{t+s+1}w-1$, which is the final run $2^m w-1$ of $\Phi(w, m)$; hence $t+s+1 = m$. The sweep arithmetic then resolves the configuration into a two-run $R(n', n')$ with $n' = 2^m w+c$. The congruence $c \equiv 3 \pmod 6$ and $c \ge 3$ were verified symbolically for every endgame cell (all $h \le 160$ exhaustively — 841/841 cells, including two independent re-audits; the full $h \le 1024$ grid adds 5049 further cells, all passing, for the ledger total 5890), with the endgame anchor form additionally matched against constructed landmark walks (216/216). The growth inequality follows: $n' - (n + 3) = c \ge 3$, i.e. $n' \ge n + 6$. □

## 7. No halting

**Theorem 7.1** (safety)**.** *No configuration on any trajectory from $P(3)$ is of the form $G(1, \cdot)$ or $G(3, \cdot)$; in particular no $G$-state has leading head $a \in \{1, 3\}$, and by Lemma 2.4 $M$ never enters the halting transition. (The master-segment part of the computational audit below is conditional on Assumption 5.10 in the standard way.)*

**Lemma 7.2** (exhaustiveness of the state classification)**.** *Every $G$-state occurring on a trajectory from $P(3)$ occurs inside one of the segment derivations of Theorem 3.4, Lemmas 5.4–5.5, the rounds of Theorem 5.3 (base table for $h \le 15$; ladder prefix rounds and master segment for $h \ge 16$), or the endgame cells of Theorem 6.1; and each such segment derivation is single-path, hence enumerates its events exactly.*

*Proof.* By the assembly recursion of §8, the trajectory from $P(3)$ is the concatenation of the segment derivations of the anchor transitions. Each segment is single-path: the prologue and cascade are deterministic algebra (Theorem 3.4, Lemma 5.4); the base rounds $h \le 15$, the ladder-prefix rounds (classes $< h$, induction), the master segments (Assumption 5.10), and the endgame cells are single-path symbolic derivations by construction/verification. A single-path derivation from a fixed initial configuration enumerates exactly the events that occur. □

*Proof of Theorem 7.1.* We split the argument into an algebraic part and a computational part.

*Algebraic part (prologue, entry, and cascade families).* Let $a$ be the leading head of a $G$-state in: (a) the prologue halving chain, where $a_i + 1 = 2^{m-i}w$ with $w \geq 3$ odd (note $w \geq 3$ since $n \equiv 3 \pmod 6$ forces $n + 3 \geq 6$); the prologue's initial head $n + 2 \geq 5$ is odd and $\neq 3$ as $n \geq 3$; (b) the halving cascade of a round with class $h$ and $t \geq h + 4$, where $a_i + 1 = 2^{h-i}\Psi$ with $\Psi = \omega + 3 \cdot 2^{t-3-h}w$; here $\omega \geq 1$ is odd and $3 \cdot 2^{t-3-h}w \geq 3 \cdot 2 \cdot 3 = 18$, so $\Psi \geq 19$ is odd; (c) a landmark $G$-head $\alpha + 3 \cdot 2^{t-3}w$ with $\alpha \equiv 1 \pmod 6$, $\alpha \geq 7$, and $3 \cdot 2^{t-3}w \geq 3 \cdot 64 \cdot 3 > 12$: the head is odd (odd + even) and $\geq 19$. In cases (a)–(c), if $a$ is odd then $a + 1 = 2^j \cdot (\text{odd} \geq 3)$ for some $j \geq 1$, while $a \in \{1, 3\}$ would require $a + 1 \in \{2, 4\} = 2^j \cdot 1$ with odd part $1 < 3$: contradiction. The even values that occur ($a = w - 1$ with $w \geq 3$ odd) are $\geq 2$ and even, hence also not in $\{1, 3\}$. So $a \notin \{1, 3\}$ throughout these families.

*Computational part (intermediate and master/endgame states).* The $G$-states inside sweeps, master segments, and endgames are exactly those enumerated by the single-path derivations of Lemma 7.2. The 2-adic audit (exact two-variable $(u, c)$ solvers) checked, over 273,654 symbolic forms — all round classes $h \leq 1024$, all endgame cells, and the full master-path form set (which, by the uniformity clause of Assumption 5.10 (Remark 5.11), is $h$-uniform) — that no form has a leading head with $a + 1$ a power of 2 (this excludes $a \in \{1, 3, 7, 15, \dots\}$ simultaneously, in particular both halting heads) or reaches such a head under any admissible continuation of its symbolic domain: 0 dangers, 0 undecided.

By Lemma 7.2 the two parts cover every $G$-state on the trajectory, and in neither part does a head in $\{1, 3\}$ occur. By Lemma 2.4, $M$ never enters the halting transition. □

## 8. Assembly and proof of the main theorem

*Proof of Theorem 1.1.* By Lemma 3.2 the run reaches $P(3)$ at step 54. Define the anchor sequence $n_0 = 3$ and, inductively, $n_{k+1} := n'$ the exit anchor of Theorem 6.1 applied to the $\Phi(w_k, m_k)$-trajectory started from $P(n_k)$ (which exists by Theorem 3.4 with $n_k + 3 = 2^{m_k}w_k$). We maintain the invariant $6 \mid n_k + 3$. It holds for $n_0 = 3$. Suppose $6 \mid n_k + 3 = 2^{m_k}w_k$; then $3 \mid w_k$, and $6 \mid 2^{m_k}w_k$ (both 2 and 3 divide it). The exit of Theorem 6.1 gives

$$n_{k+1} + 3 \;=\; 2^{m_k}w_k + (c_k + 3), \qquad 6 \mid c_k + 3 \;\; (\text{since } c_k \equiv 3 \pmod 6),$$

so $6 \mid n_{k+1} + 3$. In particular every $n_k \equiv 3 \pmod 6$ (odd), which is the domain of Theorem 3.4. Theorem 5.3 gives the complete decomposition of every transition into prologue + rounds + exit, with $n_{k+1} \geq n_k + 6$ (Theorem 6.1); the sequence is defined for all $k$; and by Theorem 7.1 no halting configuration ever occurs. Since the phase calculus of Lemma 2.3 mirrors the machine exactly (including the halting transition), $M$ never halts. □

## 9. Computational components

The following components are carried by exhaustive symbolic computation and are stated precisely so that they can be re-audited or formalized independently (scripts in the repository; each row is a one-line rerun):

| Component | Domain | Result |
|---|---|---|
| Rule-level grid (Lemma 2.3) | $a \le 20 \times b \le 10\times$ tails | 564/564 (`rulecheck`) |
| Trajectory faithfulness (Lemma 2.3) | 956 anchors ($n$=2..957) | 955 equal; $R(2,2)$ off-grammar |
| Prologue (Theorem 3.4) | $n \le 6003$; $m \le 1200$ | 1052/1052 |
| Cascade/entry (Lemmas 5.4,5.5) | $h \le 60$ | 327/327 |
| Ladder margins (Lemma 5.8) | $j < 200000$ | 199997+199992 |
| Base round law | $h \le 15$ | single-path |
| Induction step | $h = 16..1024$ | 1009/1009 |
| Master extension | $h = 1025..1.2 \cdot 10^7$ | 976 rows + 4 fresh |
| Endgame cells (Theorem 6.1) | $h \le 160$ ($\le 1024$) | 841/841 (5890 total) |
| Endgame forms vs. walks | 216 cells | 216/216 |
| 2-adic safety audit | 273654 forms | 0 dangers |
| Ledger closure (Theorem 4.3) | boundary + $2^{40}$ random | 219,398 author + 20,508 independent steps |
| Ledger orbit (Theorems 4.3, 4.6) | $10^7$ events | 0 violations; 2,908 births, 0 powers |
| Independent raw re-check | $5 \cdot 10^5$ events | 647 births, 0 powers |
| Depth-1 guard search (Proposition 4.7) | $S/L$ tails $\le 6$ | 1255 violations, one class |
| Cascade witnesses (Proposition 4.8) | 5 chains | 13/15/33/152/542 events, raw halt |

The residual in its final form. Magnitude decisions are settled unconditionally by dominance ($h \ge 82$; $h \le 81$ pointwise); $\Delta t(h) \in [2, 9]$ on all verified classes (Theorem 5.3); the halting branch is unreachable and perturbation-robust (2980/2980 single/sparse-flip counterfactuals close safely; adversarial profiles diverge but are 2-adically unrealizable and never halt). What remains is Assumption 5.10 beyond the exhaustive range ($h \le 2000$ exhaustive, sampled to $1.2 \cdot 10^7$). A finite-state reduction (`sigdyn.mjs`, `brent32.mjs`) shows this residual is exactly the continuation of the *record timetable* of the autonomous walk: first arrivals of new class depths occur at rounds climbing by a factor $\approx 4$ per level (measured schedule 17@19,024, 19@76,096, 21@304,383, 22@1,217,531, 24@4,870,122, and depth $\ge 32$ by step 1,246,751,041 of the residue kernel, which terminates at a bad residue — the record-32 event — with no cycle).

**Theorem 9.1** (periodicity excluded on the verified slice; conjecture beyond)**.** *On the* 32*-bit slice of the signature dynamics (the residual kernel of the K-slice model at* $K = 32$*): Brent cycle detection over* $1.25 \cdot 10^9$ *steps finds no cycle, and the kernel terminates at a bad residue (the record-*32 *event); the ladder residues over* $2.4 \cdot 10^6$ *positions are pairwise distinct. Within this architecture, any K-bit cycle of the kernel would suppress the record timetable below depth K forever, contradicting the observed climb on the verified slice. Consequently no finite closure of Assumption 5.10 exists at* $K \le 32$*; the corresponding statement for* $K > 32$ *— equivalently, that the full signature dynamics is not eventually periodic — is a* conjecture *(the record climb is measured, not proved, beyond the slice). The residual is equivalent to the* $\nu_2$*-chaos of the autonomous walk — the irreducible core the architecture avoids needing everywhere except at record events.*

Formalizing Lemma 2.3 and the dominance inequality in a proof assistant would raise the conditional theorem and the envelope to proof-assistant grade; eliminating Assumption 5.10 itself requires, by Theorem 9.1, mathematics beyond this architecture (or unbounded per-class computation).